Assignment 3 - Wheeled Mobile Robots

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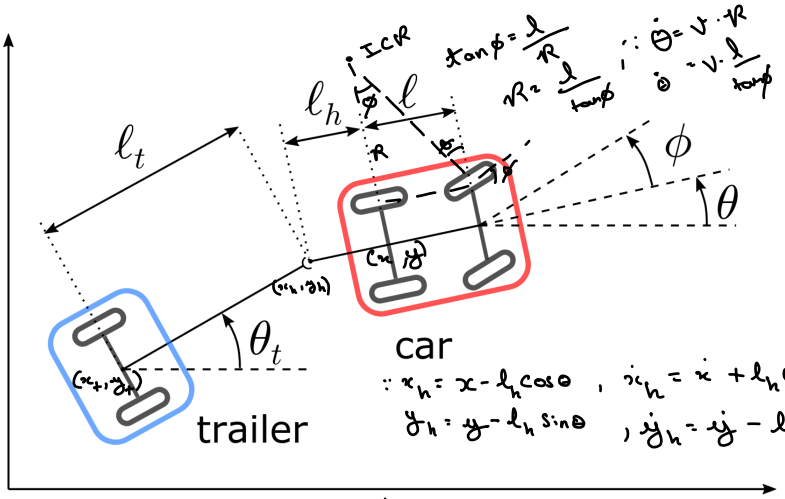
Evaluation is mostly based on how clearly you explain concepts and motivate statements, rather than computations. If you show the correct procedure but you make a mistake in the computation, it won't affect how the exercise is evaluated, as long as the procedure is properly motivated.

Exercise 1 (50 Points)

Consider the vehicle shown in figure, consisting of a rear-wheel drive car towing a trailer. The trailer is a rigid body with an axle carrying two fixed wheels, and is connected to the car through a revolute joint located at a distance ℓ_h from the rear wheel axle. Write the kinematic constraints to which the robot is subject, and derive a kinematic model for it.



$$\tan\phi = \frac{l}{R}, \quad R = \frac{l}{\tan\phi} \quad \therefore \dot{\theta} = v \cdot R$$
$$\dot{\theta} = v \cdot \frac{l}{\tan\phi}$$

$$\therefore x_h = x - l_h\cos\theta, \quad \dot{x}_h = \dot{x} + l_h\dot{\theta}\sin\theta$$
$$y_h = y - l_h\sin\theta, \quad \dot{y}_h = \dot{y} - l_h\dot{\theta}\cos\theta \quad \text{— } 1$$

$$x_t = x_h - l_t\cos\theta_t, \quad \dot{x}_t = \dot{x}_h + l_t\dot{\theta}_t\sin\theta_t = \dot{x} + l_h\dot{\theta}\sin\theta + l_t\dot{\theta}_t\sin\theta_t$$
$$y_t = y_h - l_t\sin\theta_t, \quad \dot{y}_t = \underset{\text{from } 1}{\dot{y}_h} - l_t\dot{\theta}_t\cos\theta_t = \dot{y} - l_h\dot{\theta}\cos\theta - l_t\dot{\theta}_t\cos\theta_t$$

Exercise 2 (50 Points)

Fill in the script `hw03.py`. It's about the bicycle dynamics. You will have to simulate the free evolution of the neutral bicycle model you've seen in lecture 1. Refer to slides 109-118 of lecture 1. What are the state variables? What do you expect from the bicycle evolution? Why do you expect it staying upright or falling down? Provide the answers either in the code as a comment or in the sheet you hand in for exercise 1.